

Muhammad Ammaz

Agentic AI Engineer & Full-Stack Developer

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Professional Summary

Agentic AI Engineer and Full-Stack Developer with hands-on production experience building LLM-powered systems using LangChain, OpenAI Agents SDK, and Retrieval-Augmented Generation (RAG) pipelines. Currently developing a Crime Intelligence Platform and Social Media Listening Agent at GCS Pvt Ltd — both deployed in real client environments. Experienced in MERN Stack (MongoDB, Express.js, React, Node.js), microservices architecture, Redis caching, and Docker-based deployment. Proven ability to deliver end-to-end AI solutions from data ingestion to intelligent insight generation.

Technical Skills

Programming Languages: JavaScript, TypeScript, Python

Frontend: React.js, Next.js, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, JWT Authentication

AI & Machine Learning: LangChain, LangGraph, OpenAI Agents SDK, ChromaDB, Advanced RAG Pipelines, Agentic Workflows, Multi-Agent Systems, Sentiment Analysis, Deep Learning (VGG16)

Databases: MongoDB, Microsoft SQL Server, Redis

DevOps & Tools: Docker, Git, Postman, SonarQube

Architecture & Practices: Microservices, Event-Driven Architecture, Agile/SDLC, OOP, Unit Testing (Jest, Mocha)

Work Experience

Agentic AI Engineer

GCS Pvt Ltd

January 2026 – Present

Karachi, Pakistan

- *Note: Source code confidential — developed under NDA for client deployment. Worked within an AI engineering team delivering enterprise-grade LLM applications for government and enterprise clients.*
- **Crime Intelligence RAG System:** Architected and deployed a production-grade AI-powered crime intelligence platform using **LangGraph** and LangChain with OpenAI APIs, enabling law enforcement agencies to extract actionable insights from large-scale FIR records and historical criminal datasets.
- Engineered an **Advanced RAG pipeline** incorporating query rewriting and agent-driven dynamic retrieval — agents reformulate and route queries based on request context using LangGraph state transitions — delivering more accurate intelligence summaries compared to naive RAG approaches.
- Implemented **LangGraph-based agentic workflows** with stateful reasoning graphs, dynamically routing retrieval workflows based on query context via LangGraph routing logic, and validating intermediate steps before generating final intelligence reports.
- Designed a **multi-agent architecture** using LangGraph where specialized agents handle distinct tasks — evidence retrieval, pattern analysis, crime hotspot analysis, and report generation — coordinated through a supervisor agent graph.
- Built a data ingestion portal to securely collect, validate, and manage FIR reports and criminal records, feeding structured and unstructured data into **ChromaDB** vector store for semantic retrieval.
- Architected using microservices and event-driven design with **Redis caching** on retrieved context to reduce repeated vector search overhead and improve response latency under high concurrency.
- Containerized the full stack using **Docker** and deployed on an on-premise server with secure port-sharing, meeting strict client data residency and security requirements.
- **Social Media Listening Agent:** Built an AI-based monitoring system to track user sentiment across multiple platforms in real time.
- Developed autonomous agents to collect posts, comments, and engagement data continuously without manual intervention.

- Implemented NLP-based sentiment analysis and trend analysis to surface public reactions and emerging discussion patterns across monitored topics.
- Engineered an automated response module enabling the agent to contextually reply to comments and posts.
- **HireFlow – AI-Powered HR Automation:** Designed and developed a full-stack AI-powered HR automation platform using LangChain and OpenAI APIs to eliminate repetitive HR tasks and accelerate the end-to-end recruitment lifecycle.
- Built an intelligent **resume screening engine** that parses and evaluates candidate CVs against job descriptions using semantic similarity and LLM-based scoring, significantly reducing manual screening time for HR teams.
- Implemented an **automated candidate ranking system** that scores applicants across multiple criteria (skills match, experience, education) and generates a prioritized shortlist for HR teams.
- Developed an **AI Interview Module** that auto-generates role-specific interview questions, conducts asynchronous text-based interviews with candidates, and produces structured evaluation reports for hiring managers.
- Integrated **email automation** to send interview invitations, rejection notices, and status updates to candidates, reducing HR communication overhead significantly.
- Architected the backend with **Node.js and Express.js** with a **React.js** dashboard for HR managers to manage job postings, review ranked candidates, and track hiring pipeline status in real time.
- Leveraged **ChromaDB** as a vector store for semantic search across resume databases, enabling fast and accurate candidate retrieval even across large applicant pools.

MERN Stack Intern

October 2025 – December 2025

10Pearls

Karachi, Pakistan

- Built a production-grade Full-Stack Notes application with user registration, authentication, and full CRUD operations using the MERN stack.
- Achieved comprehensive test coverage with 18 backend and 6 frontend unit tests using Jest.
- Integrated SonarQube for static code analysis and resolved all flagged quality issues.
- Implemented secure RESTful JWT authentication with stateless session management and role-based access control.
- Participated in code reviews with senior developers, improving code architecture and best practices.

MERN Stack Intern

August 2025 – September 2025

RealTech Solutions

Karachi, Pakistan

- Updated and maintained modules in a large-scale ERP system on Node.js and Express.js.
- Authored complex SQL queries against Microsoft SQL Server for management reporting dashboards.
- Designed dashboard interfaces for sales performance, inventory, and operational KPI visualization.
- Optimized API calls and queries, achieving a **20% reduction in page load times**.
- Diagnosed and resolved user-reported bugs with thorough pre-deployment testing.

Projects

Chest Guard: Medical Image Analysis System | React, Node.js, Express, MongoDB, Python, VGG16 Final Year Project

- Architected a full-stack web app to analyze 4,500+ chest X-rays, detecting Tuberculosis and Pneumonia via a trained deep learning model.
- Designed a responsive React + Tailwind CSS frontend for image uploads and diagnostic prediction display.
- Developed Node.js/Express APIs to receive images, forward to VGG16 model, and return structured predictions.
- Trained and fine-tuned VGG16 CNN in Python for multi-class chest disease classification.
- Persisted user data and prediction history in MongoDB for physician review and patient tracking.

E-Commerce Marketplace | Next.js, Sanity CMS, Clerk Auth, Stripe

72-Hour Hackathon

- Delivered a complete online marketplace with product pages, shopping cart, and checkout in 72 hours.
- Integrated Sanity CMS for headless product catalog management.
- Implemented Clerk for user registration, login, and session management.
- Integrated Stripe API for secure test payment processing.
- Built an admin dashboard for real-time order viewing and fulfillment status updates.

Education

Sir Syed University of Engineering and Technology
Bachelor of Science in Software Engineering (BSSE)

Karachi, Pakistan
Graduated: 2025